In [1]:
```
# importing Major libraries

import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

import warnings
warnings.filterwarnings('ignore')
```

In [2]:
```
df = pd.read_csv('human_activity_recognition.csv').drop('subject',axis=1)
```

In [3]:
```
df.head()
```

Out[3]:

	tBodyAcc-mean()-X	tBodyAcc-mean()-Y	tBodyAcc-mean()-Z	tBodyAcc-std()-X	tBodyAcc-std()-Y	tBodyAcc-std()-Z	tBodyAcc mad()-
0	0.288585	-0.020294	-0.132905	-0.995279	-0.983111	-0.913526	-0.99511
1	0.278419	-0.016411	-0.123520	-0.998245	-0.975300	-0.960322	-0.99880
2	0.279653	-0.019467	-0.113462	-0.995380	-0.967187	-0.978944	-0.99652
3	0.279174	-0.026201	-0.123283	-0.996091	-0.983403	-0.990675	-0.99709
4	0.276629	-0.016570	-0.115362	-0.998139	-0.980817	-0.990482	-0.99832

5 rows × 562 columns

In [4]:
```
# df.subject.describe()
df.shape
```

Out[4]:
```
(7352, 562)
```

In [5]:
```
# y
import plotly.express as px
px.pie(values=df.Activity.value_counts(),names=df.Activity.value_counts().inde

df.Activity.value_counts()
```

Out[5]:
```
Activity
LAYING                1407
STANDING              1374
SITTING               1286
WALKING               1226
WALKING_UPSTAIRS      1073
WALKING_DOWNSTAIRS     986
Name: count, dtype: int64
```

In [6]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score,confusion_matrix
```

In [7]:
```
lr = LogisticRegression(multi_class='ovr')
lr
```

Out[7]:
```
▾ LogisticRegression
LogisticRegression(multi_class='ovr')
```

In [8]:
```
x = df.iloc[:,0:-1]
y = df.Activity

x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2,random_stat
```

In [9]:
```
print(x_train.shape,x_test.shape,y_train.shape,y_test.shape)
```

```
(5881, 561) (1471, 561) (5881,) (1471,)
```

In [10]:
```
lr.fit(x_train,y_train)
```

Out[10]:
```
▾ LogisticRegression
LogisticRegression(multi_class='ovr')
```

In [11]:
```
y_pred = lr.predict(x_test)

accuracy_score(y_test,y_pred)
```

Out[11]:
```
0.982324949014276
```

In [12]:
```
# manual filteration

df.duplicated().sum()
```

Out[12]:
```
np.int64(0)
```

In [13]:
```
df.T.duplicated().sum()
```

Out[13]:
```
np.int64(21)
```

In [14]:
```
df = df.T.drop_duplicates().T
```

In [15]:
```
# variance threshold

# 0.01 - 0.05
```

In [16]:
```
x = df.iloc[:,0:-1]
y = df.Activity

x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2,random_stat
print(x_train.shape,x_test.shape,y_train.shape,y_test.shape)
```

```
(5881, 540) (1471, 540) (5881,) (1471,)
```

In [17]:
```
# manual way
l1 = df.drop('Activity',axis=1).var()[df.drop('Activity',axis=1).var()<0.05].i
```

In [18]:
```
# df.drop(l1,axis=1)
```

In [19]:
```
# automatic way
```

```
from sklearn.feature_selection import VarianceThreshold

vt = VarianceThreshold(threshold=0.05)

vt
```

Out[19]:

```
▾ VarianceThreshold
VarianceThreshold(threshold=0.05)
```

In [20]:

```
x_train_vt = pd.DataFrame(vt.fit_transform(x_train),columns=vt.get_feature_nam
x_test_vt = pd.DataFrame(vt.transform(x_test),columns=vt.get_feature_names_out
```

In [21]:

```
print(x_test.shape[1])
x_test_vt.shape[1]
```

```
540
```

Out[21]:

```
349
```

In [22]:

```
# multicollinearity

# sns.heatmap(x_train_vt.corr(),annot=True)
```

In [23]:

```
x_train_vt
```

Out[23]:

	tBodyAcc-std()-X	tBodyAcc-std()-Y	tBodyAcc-std()-Z	tBodyAcc-mad()-X	tBodyAcc-mad()-Y	tBodyAcc-mad()-Z	tBody ma
0	-0.994425	-0.994873	-0.994886	-0.994939	-0.993994	-0.99545	-0.93
1	-0.326331	0.069663	-0.224321	-0.343326	0.039623	-0.256327	-0.31
2	-0.02622	-0.032163	0.393109	-0.118256	-0.030279	0.432861	0.37
3	-0.981092	-0.901124	-0.960423	-0.984417	-0.901405	-0.965788	-0.92
4	-0.99738	-0.983893	-0.984482	-0.997331	-0.985196	-0.983768	-0.94
...	...	...	...	...	...	...	
5876	-0.555352	-0.104055	-0.438064	-0.57253	-0.112149	-0.429688	-0.47
5877	-0.290043	-0.212102	-0.469731	-0.307317	-0.209558	-0.528635	-0.28
5878	-0.627198	-0.216566	-0.424764	-0.648666	-0.253814	-0.417569	-0.54
5879	-0.994825	-0.985314	-0.965857	-0.99517	-0.984285	-0.963293	-0.93
5880	-0.368655	-0.142631	-0.15125	-0.426026	-0.130656	-0.149079	-0.21

5881 rows × 349 columns

In [24]:

```
x_train_vt
```

Out[24]:

	tBodyAcc-std()-X	tBodyAcc-std()-Y	tBodyAcc-std()-Z	tBodyAcc-mad()-X	tBodyAcc-mad()-Y	tBodyAcc-mad()-Z	tBody ma
0	-0.994425	-0.994873	-0.994886	-0.994939	-0.993994	-0.99545	-0.93
1	-0.326331	0.069663	-0.224321	-0.343326	0.039623	-0.256327	-0.31
2	-0.02622	-0.032163	0.393109	-0.118256	-0.030279	0.432861	0.37
3	-0.981092	-0.901124	-0.960423	-0.984417	-0.901405	-0.965788	-0.92
4	-0.99738	-0.983893	-0.984482	-0.997331	-0.985196	-0.983768	-0.94
...	...	...	...	...	...	...	
5876	-0.555352	-0.104055	-0.438064	-0.57253	-0.112149	-0.429688	-0.47
5877	-0.290043	-0.212102	-0.469731	-0.307317	-0.209558	-0.528635	-0.28
5878	-0.627198	-0.216566	-0.424764	-0.648666	-0.253814	-0.417569	-0.54
5879	-0.994825	-0.985314	-0.965857	-0.99517	-0.984285	-0.963293	-0.93
5880	-0.368655	-0.142631	-0.15125	-0.426026	-0.130656	-0.149079	-0.21

5881 rows × 349 columns

In [25]:
```
def remove_multicollinearity(df, threshold=0.9):
    col_corr = set()
    corr_matrix = df.corr()
    for i in range(len(corr_matrix.columns)):
        for j in range(i):
            if abs(corr_matrix.iloc[i, j]) > threshold:
                colname = corr_matrix.columns[i]
                col_corr.add(colname)
    return df.drop(columns=list(col_corr))
```

In [26]:
```
x_train_vt_no_mc = remove_multicollinearity(x_train_vt)
x_test_vt_no_mc = remove_multicollinearity(x_test_vt)
```

In [27]:
```
print(f"Shape of x_train_vt before removing multicollinearity: {x_train_vt.sha
print(f"Shape of x_train_vt after removing multicollinearity: {x_train_vt_no_m
print(f"Shape of x_test_vt before removing multicollinearity: {x_test_vt.shape
print(f"Shape of x_test_vt after removing multicollinearity: {x_test_vt_no_mc.
```

```
Shape of x_train_vt before removing multicollinearity: (5881, 349)
Shape of x_train_vt after removing multicollinearity: (5881, 112)
Shape of x_test_vt before removing multicollinearity: (1471, 349)
Shape of x_test_vt after removing multicollinearity: (1471, 112)
```

In [28]:
```
# anova

from sklearn.feature_selection import f_classif
from sklearn.feature_selection import SelectKBest

selector = SelectKBest(f_classif, k=100)
```

```
anova = f_classif(x_train_vt,y_train)

selector
```

Out[28]:

```
▾ SelectKBest
SelectKBest(k=100)
```

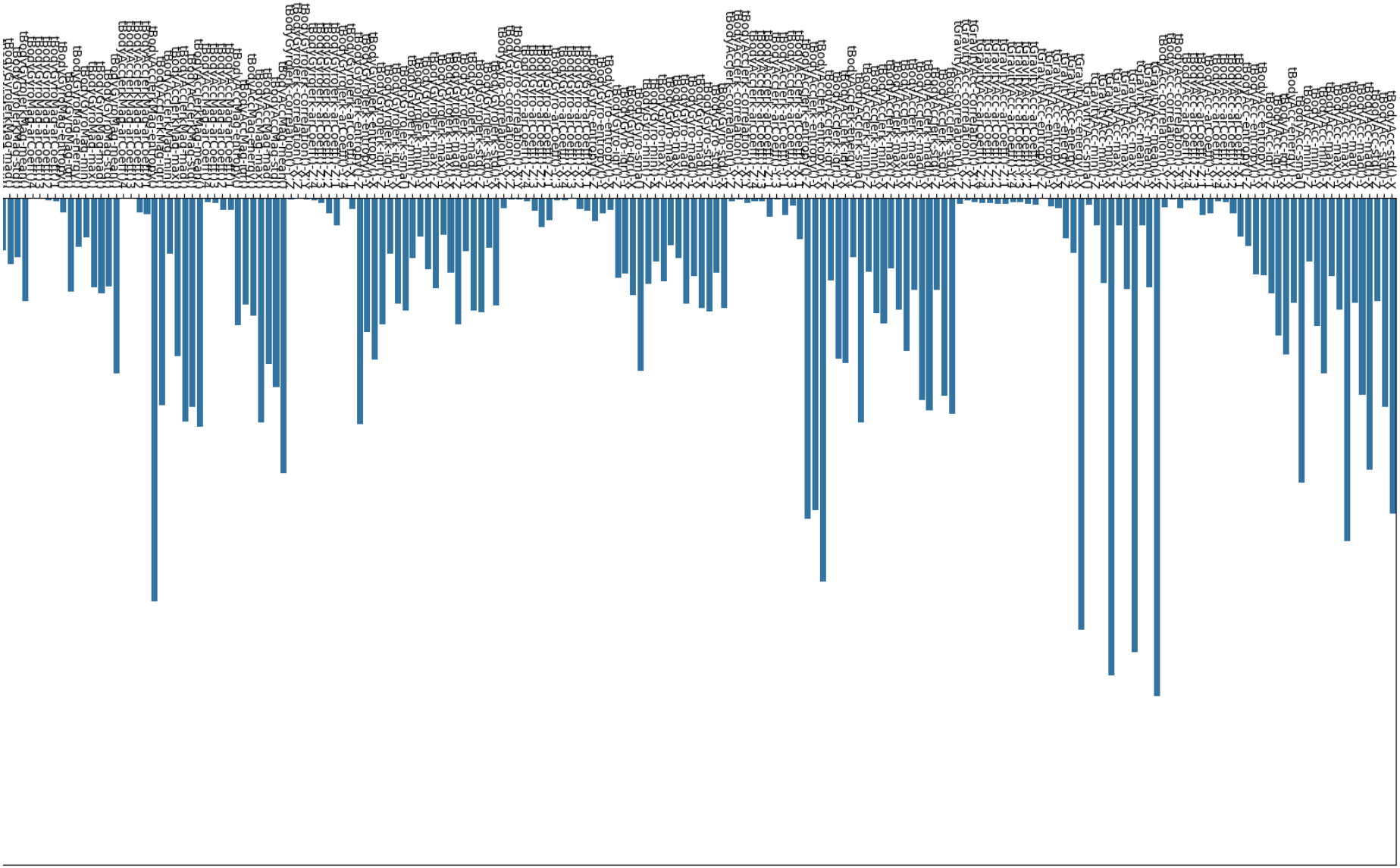
In [30]:

```
plt.figure(figsize=(10,40))
sns.barplot(y=x_train_vt.columns,x=anova[0])
```

Out[30]:

```
<Axes: ylabel='None'>
```

None



In [34]:
```
from sklearn.feature_selection import SelectKBest, f_classif

X = df.iloc[:, :-1]
y = df.iloc[:, -1]

selector = SelectKBest(score_func=f_classif, k=5)

X_new = selector.fit_transform(X, y)
```

In [35]:
```
l1 = selector.get_feature_names_out()
```

In [39]:
```
selector = SelectKBest(f_classif, k=80)
# anova = f_classif(x_train_vt,y_train)
```

In [40]:
```
from sklearn.feature_selection import SelectKBest, f_classif
import pandas as pd

selector = SelectKBest(score_func=f_classif, k=5)

X_new = selector.fit_transform(x_train_vt, y_train)

selected_cols = x_train_vt.columns[selector.get_support()]

df_new = pd.DataFrame(X_new, columns=selected_cols)

print(df_new.head())
```

```
   tGravityAcc-mean()-X tGravityAcc-max()-X tGravityAcc-min()-X  \
0             -0.208633           -0.277682           -0.163251
1              0.782327            0.717677            0.793836
2              0.954247            0.895073            0.938414
3              0.947747            0.876104             0.96641
4              0.883139            0.809617             0.90257

   fBodyAccJerk-entropy()-X fBodyAccJerk-entropy()-Y
0                      -1.0                      -1.0
1                  0.277542                  0.486605
2                  0.662595                  0.400274
3                 -0.943699                  -0.93912
4                      -1.0                      -1.0
```

In [41]:
```
pd.DataFrame(selector.fit_transform(x_train_vt,y_train),columns=l1)
```

Out[41]:

	tGravityAcc-mean()-X	tGravityAcc-max()-X	tGravityAcc-min()-X	fBodyAccJerk-entropy()-X	fBodyAccJerk-entropy()-Y
0	-0.208633	-0.277682	-0.163251	-1.0	-1.0
1	0.782327	0.717677	0.793836	0.277542	0.486605
2	0.954247	0.895073	0.938414	0.662595	0.400274
3	0.947747	0.876104	0.96641	-0.943699	-0.93912
4	0.883139	0.809617	0.90257	-1.0	-1.0
...	...	...	...	...	...
5876	0.913096	0.847183	0.929265	0.362592	0.445242
5877	0.888937	0.829889	0.895652	0.197617	0.363821
5878	0.968026	0.898462	0.980335	0.227114	0.370431
5879	0.956321	0.883129	0.975233	-1.0	-1.0
5880	0.962919	0.900097	0.971153	0.382117	0.495643

5881 rows × 5 columns

In [ ]: